# The pair partition in the paired equality case

Human Compression project: a component of the L044 recovery

Retain the setting of *Neighbours of a square in a paired selection*: a transitive string action on $n = 2r > 8$ vertices, with two selected eligible edges of every colour and a connected selection whose unique cycle is an alternating square. A square is admissible if it is the unique cycle of such a selection. We use the acquired four-root-tree lemma, the choice containing all actual neighbours of a fixed square, the square transport lemma, and the bound three for the actual degree of each vertex of any admissible square. Eligibility always refers to the actual omission group $G_i = \langle \rho_j : j \neq i \rangle$.

The source ladder strategy is due to Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for $A_n$*, arXiv:1605.09173v1, Propositions 4.16–4.18. We give the selection and continuation details explicitly. In particular, a slide below changes only the rung edges: it does not silently replace an edge of another colour farther along the selection.

**Lemma 1** (Squares with no common middle label)**.** *If an admissible square has colours $a, b$ with $|a - b| \neq 2$, every connected paired selection having that square as its unique cycle already contains every actual edge incident with the square.*

*Proof.* Use the four root trees. An unselected actual $k$-edge from a square vertex $p$ ends in $T_p$, and the tree path to its other endpoint starts with a colour $\ell \neq k$. The path and the unselected edge form a cycle. Eligibility forces both selected $\ell$-edges onto that path. Transporting its first edge by any square mark commuting with $\rho_\ell$ would require a selected $\ell$-edge in a different root tree. Thus $\ell$ is adjacent to both $a$ and $b$, forcing $|a - b| = 2$, contrary to hypothesis. □

**Lemma 2** (Continuation along pairs)**.** *Suppose the current selected square has rung colour $a$ and longitudinal colour $c$, with $|a - c| \geq 3$. Write its two $a$-pairs as $B, F$, joined by the two $c$-edges. Suppose each vertex of $B$ has a third actual edge of colour $d = c - \epsilon$, where $\epsilon \in \{-1, 1\}$ and these are the two selected $d$-edges. Then every actual edge leaving the square at $F$ has colour $c + \epsilon$.*

*If the difference from $a$ is increasing in direction $\epsilon$, such an edge has a parallel mate, both are selected, and moving the square to these new endpoints changes only one selected $a$-edge. All other selected edges are retained. Whenever all previous vertices other than $F$ admit no selected exit, and the new endpoints are outside those vertices, this extends the selected ladder by one pair. If there is no exit at $F$, that ladder is the whole connected selection.*

*Proof.* Let $e$ be the colour of a further edge at a vertex of $F$. If it commuted with $c$, transport by $\rho_c$ would give an edge of colour $e$ at the corresponding vertex of $B$. That vertex already has its three actual edges of colours $a, c, d$. Since $e \notin \{a, c\}$ and $d$ is adjacent to $c$, none of these can be the commuting colour $e$. Hence $e$ must be $c - 1$ or $c + 1$. Lemma 1 makes this edge selected. The two selected $d$-edges already meet $B$, so $e \neq d$, leaving exactly $e = c + \epsilon$.

In the stated direction $e$ commutes with $a$. Its parallel edge meets the other vertex of $F$ and is also selected by Lemma 1. The two new endpoints lie outside the square, in its two different front root trees. Replace the $a$-edge on $B$ by the transported $a$-edge on these endpoints. The acquired square transport proof shows that this leaves a connected paired selection with the new square as its unique cycle. No other edge is changed. The final assertions follow from connectedness: a selected path from the displayed ladder to any other vertex would have to leave through its current front. □

**Lemma 3** (A square whose labels differ by two)**.** *There exists an admissible square with colours* $a, a+2$*.*

*Proof.* If an admissible square has consecutive colours, choose a selection containing all its actual neighbours. Connectedness and $n > 4$ give an edge leaving it. The root-tree lemma makes its colour the outer neighbour of one square colour. It commutes with the other, so square transport gives an admissible square with label difference two.

Suppose, for contradiction, that no difference-two square exists. Choose an admissible square with minimum label difference, say colours $a < b$. The preceding paragraph implies $b - a \geq 3$. An edge leaving the square is adjacent to a square colour. An inner attachment of colour $a+1$ or $b-1$ would give an admissible square of smaller difference. Thus every attachment points outward. Reverse the full string if necessary so that one has colour $c = a - 1$.

Choose the selection containing all neighbours of this square. The two parallel $c$-edges leave one $b$-pair, denoted $R_1$, and reach a new pair $R_2$. Let $R_0$ be the other $b$-pair of the original square. Both vertices of $R_1$ now have actual degree three, with colours $b, a, c$. Any additional edge at $R_0$ must fail to commute with $a$, since otherwise transport to $R_1$ would exceed its degree. Its colour is therefore $a-1$ or $a+1$. The former already has its two selected edges at $R_1$, and all neighbours of the original square are selected. The latter would give a smaller admissible label difference. Thus $R_0$ has actual degree exactly two.

Slide the square to $R_1, R_2$, with rung $b$ and longitudinal colour $c = a - 1$. Apply Lemma 2 with $\epsilon = -1$. Every successive extension is a parallel pair of edges whose labels decrease by one. The rear pair $R_0$ is closed; every subsequent interior vertex has its three actual edges. New endpoints cannot be previously constructed vertices, which already have their complete degree or are in the closed rear pair. Thus the construction continues along distinct pairs or stops at a closed front. It must stop, since the labels cannot decrease indefinitely. All full-graph exits have then been excluded, so transitivity makes these pairs all of $\Omega$. But the only colours used on them are $b, a, a-1, \ldots$; a colour strictly between $a$ and $b$ fixes every vertex. This is incompatible with having two eligible edges of every colour. The contradiction proves the lemma. □

**Lemma 4** (The two middle edges)**.** *Choose a selection containing all actual neighbours of an admissible square with colours* $a, b = a + 2$*, and put* $k = a + 1$*. Both selected* $k$*-edges meet the square, at two adjacent vertices. Their other endpoints have degree one in the selected graph.*

*Proof.* First suppose no $k$-edge meets the square. Some selected edge leaves it, by connectedness. Its colour is $a - 1$ or $b + 1$. Reverse labels if needed to choose $c = b + 1$. It and its parallel mate join one $a$-pair to a new pair. The other $a$-pair of the square is closed: an additional edge there must fail to commute with $b$, by transport to the now saturated pair, so has colour $b - 1 = k$ or $b + 1 = c$. The first is absent by hypothesis and the second already uses its two selected edges at the saturated pair.

After sliding to the new square with colours $a, c$, its label difference is three. Lemma 2, now with $\epsilon = 1$, continues along increasing longitudinal labels. The closed rear and saturated interior vertices force every new pair to be disjoint from the preceding ones. The process ends because there are finitely many labels, and the resulting full component omits colour $k$. As in Lemma 3, transitivity gives a contradiction. Thus some selected $k$-edge $pu$ leaves the square.

The vertex $u$ is a selected leaf. Indeed all selected $a$- and $b$-edges already lie in the square, and the involution $\rho_k$ already pairs $u$ with $p$. Any other selected edge at $u$ would have a colour commuting with $k$. Transport would give a fourth actual edge at $p$, whose three edges have colours $a, b, k$. This is impossible. The same reasoning applies to every selected middle edge meeting the square.

Write $q = p^{\rho_a}$, $s = p^{\rho_b}$ and $t = p^{\rho_a \rho_b}$. Suppose the second selected $k$-edge does not meet the square. Since $u$ is a selected leaf, connectedness forces an outer-colour attachment at one of $q, s, t$. Reverse labels if necessary so that its colour is $c = b + 1$. As it commutes with $a$, and $p$ is already saturated, its two parallel edges must leave $s, t$. Both of these vertices are now

saturated. The vertex $q$ is closed: an attachment there would be of colour $a-1$ or $b+1$, and commuting transport by $b$ or $a$, respectively, would give a fourth edge at $t$ or $p$. There is no actual $k$-edge at $q$ under the current assumption, since all actual square neighbours are selected.

Slide the rung colour $a$ to the new pair. The new square has label difference three. Continue outwards by Lemma 2. Every slide changes only an $a$-edge and retains $pu$ and the other selected $k$-edge. The leaf $u$ remains a selected leaf: the selected $a$-edges lie on the current square, the selected $b$-edges are the original two longitudinal edges, and every other possible colour except $k$ commutes with $k$ and fixes $p$, hence fixes $u$. All previous nonleaf vertices are closed or saturated. Each new longitudinal colour is at least $b+1 = k+2$, so it also fixes $u$ by this commutation. Consequently new endpoints cannot be old vertices, and any selected exit is through the current front.

At each front the only possible new colour is the next increasing longitudinal colour. No second $k$-edge ever enters the displayed ladder. When the process stops, connectedness says this ladder and its leaf comprise the entire selection, although it contains just one selected $k$-edge. This is a contradiction.

The second middle edge therefore meets another square vertex and ends in another selected leaf. It cannot meet $t$, diagonally opposite $p$. In that case $p, t$ are saturated and both leaves have selected degree one. Any selected exit through $q$ or $s$ must have outer colour $a-1$ or $b+1$; commuting transport to $p$ or $t$ again contradicts degree three. The whole connected selection would then have only six vertices. Since $n > 8$, the two middle edges instead meet adjacent square vertices. □

**Theorem 5** (The invariant pair partition)**.** *The group $G$ preserves a partition of $\Omega$ into $r$ pairs. In particular a primitive action of degree $n = 2r > 8$ cannot have the stated paired equality selection.*

*Proof.* Use Lemmas 3 and 4. Reverse labels if necessary so the middle edges of colour $k = a+1$ meet an $a$-pair $R_0$ of the square of colours $a, b = a+2$. Let $L$ be the pair of selected leaves, and let $R_1$ be the other $a$-pair. The two vertices of $R_0$ have degree three. Since $n > 6$ and the leaves have selected degree one, some selected edge must leave $R_1$. Its colour cannot be $a-1$, which commutes with $b$ and would produce an edge at the saturated pair $R_0$. It is therefore $b+1$. Its parallel mate gives a new pair $R_2$.

Slide the $a$-rung to the new pair and continue by Lemma 2. Both leaves remain selected leaves by the same argument as in Lemma 4: each slide changes only the rung colour, and the only possibly noncommuting colours $a, b$ have their selected edges elsewhere. All newly encountered colours commute with $k$ and fix $R_0$, hence fix $L$. The remaining old vertices are saturated. Thus the construction uses distinct pairs and, by connectedness, ends only when it includes all vertices of $Q$.

The selected colours in this completed graph are exactly the consecutive interval $a, a+1, \ldots, j$, where $j$ is the last longitudinal colour. Every label $0, \ldots, r-1$ has two selected edges, so $a = 0$ and $j = r-1$. We obtain pairs

$$L, R_0, R_1, \ldots, R_{r-2}.$$

There are two edges of colour 1 between $L$ and $R_0$, and two edges of colour $i$ between $R_{i-2}$ and $R_{i-1}$ for $2 \le i \le r-1$. Each $R_j$ is an actual 0-edge, as supplied by successive commuting transports. Both vertices of every $R_j$ with $j < r-2$ have their three actual edges.

The terminal pair $R_{r-2}$ has exactly its actual edges of colours $0, r-1$. Indeed apply the first paragraph of Lemma 2 to the final square: any additional edge would have colour $r-2$ or $r$. The latter is outside the label set. The former already has its two selected edges incident with the preceding pair, and all actual neighbours of the final square are selected, by Lemma 1. Here $r \ge 5$, so its label difference is at least four, as required.

There are no vertices left outside these pairs. Since every vertex other than the two in $L$ has all its actual edges accounted for, the only possible additional actual edge joins the two leaves. Such an edge must have colour 0: a colour different from 0 would close the path from one leaf to the other through its two 1-edges and the original 0-rung on $R_0$, making a cycle with just one eligible 0-edge. The original rung is eligible whether or not it is retained in the final selection. Thus $\rho_0$ interchanges each pair $R_j$ and either fixes or interchanges the two leaves. Each $\rho_i$, $i \geq 1$, swaps the corresponding two adjacent pairs by its displayed two edges and fixes the remaining vertices. Every generator therefore preserves the stated partition. □

**Exact scope.** The acquired normal-form and equality-connectedness results, together with Theorem 5, exclude $2r = n$ for primitive actions with a paired selection and $n > 8$. The paired-cut count already gives $2r \leq n$, so in a primitive action of degree $n \geq 12$ it gives $r \leq \lfloor (n-1)/2 \rfloor$. This recovers the needed all-paired branch of L044. It does not classify every imprimitive equality group or supply the separate general transitive-imprimitive omission branch, induction bases, small maxima or lower existence. Those remain part of the complete principal-source contract. No whole-source or new open-conjecture credit is asserted here.